

Edit & Submit

Dungeon master

Hide >

CPU TIME LIMIT
1 second

MEMORY LIMIT
1024 MB

2.8 DIFFICULTY
Medium

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 Sample data files

SOURCE & LICENSE
University of Ulm
Local Contest
1997

The input file consists of a number of dungeons. Each dungeon description starts with a line containing three integers L , R and C (all limited to 30 in size). L is the number of levels making up the dungeon. R and C are the number of rows and columns making up the plan of each level.

Then there will follow L blocks of R lines each containing C characters. Each character describes one cell of the dungeon. A cell full of rock is indicated by a '#' and empty cells are represented by a '.'. Your starting position is indicated by 'S' and the exit by the letter 'E'. There's a single blank line after each level. Input is terminated by three zeroes for L , R and C .

Output Specification

Each maze generates one line of output. If it is possible to reach the exit, print a line of the form

Escaped in <x> minute(s).

where <x> is replaced by the shortest time it takes to escape. If it is not possible to escape, print the line

Trapped!

Sample Input 1

```
3 4 5
S....
.###.
.##..
###.#

#####
#####
##.##
##...

#####
#####
#.###
####E

1 3 3
S##
#E#
###

0 0 0
```

Sample Output 1

```
Escaped in 11 minute(s).
Trapped!
```

